Randy Dyer

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WORK EXPERIENCE

Software Engineer, Space Age Electronics Jan 2025 – Present

- Developed touchscreen annunciator for fire and life safety using Electron.js to integrate with fire panels and visualize real-time device statuses via serial connection.
- Built admin version of annunciator with drag-and-drop editor that generates configuration files defining UI layout and device point-to-point connections, enabling rapid deployment for custom buildings.
- Presented live product demos to executive stakeholders and investors.
- Built 3D configurator integrated into Epicor ERP system for positioning electrical components in custom fire safety cabinets and automatically generating AutoCAD drawings, streamlining the design workflow.
- Designed and implemented Microsoft Power Automate workflow to route engineering requests to Microsoft Planner with automated email notifications for submissions and completions.
- Created Power BI dashboard tracking key performance metrics including average task turnaround time.

App Developer and Web Designer, Alton Therapeutic Massage Jul 2024 – Present

- Independently created and implemented the system design for the company website and app from the ground up
- Increased client inquiries by 70% through the creation of the company website
- Securely stored sensitive data (PHI) in Amazon RDS with encryption and secured with Cognito for user authentication
- Centralized client information including contact details, health records, symptoms, and appointment notes in a full-stack React web application hosted on AWS Amplify
- Reduced new client appointment times by creating a digital intake form to fill out before visits

Mobile Developer, Student Success Application Jul 2023 – Nov 2023

- Developed a virtual handbook for WPI students that covers topics such as Academic Support, Career Development, Culture & Diversity and Wellness resources
- Created user interface components with React Native for pages within the handbook

PROJECTS

Beyond the Visual Application, Major Qualifying Project Aug 2023 – Mar 2024

- Developer for the Worcester Art Museum's mobile app (built with React Native, mongoDB and Firebase)
- Developed and tested an indoor positional system (IPS) with data from Minew BLE beacons placed in exhibits and utilizing trilateration techniques to locate and navigate individuals
- Added accessibility labels to UI for screen readers and an audio player for art piece descriptions
- Conducted user testing with both visually impaired and non-visually impaired individuals on all app features

Hospital Service Application, Brigham and Women's Hospital Mar 2023 – May 2023

- Led back-end development for a team of 10, managing the AWS PostgreSQL RDS and associated Java code
- Facilitated the integration of features with the front-end (submitting and receiving service requests, hospital navigation, and admin features such as map editing and employee updates)
- Utilized Agile Methodology and Software Design patterns throughout the creation of the app

Water Conservation Outreach, Board of Water supply Jan 2023 – Feb 2023

- Collected and analyzed 168 survey responses across two studies to identify educational content and determine optimal platforms for reaching individuals across age groups
- Created a memory matching game with HTML, CSS, and JavaScript to teach children water conservation tips
- Conducted user testing through local elementary schools and obtained survey feedback from teachers

SKILLS

Programming Languages: Typescript, JavaScript, Java, Python, SQL, noSQL, C, C++

Development: Agile, Git/GitHub, Node.js, Electron.js, AWS (EC2, RDS, Lamda, VPC, Cognito, Amplify, API Gateway), REST API, PostgreSQL, mongoDB, Pandas, NumPy, React/React Native, macOS, Windows, Linux/Ubuntu

Software/Tools: Jira, Figma, VSCode, AutoCAD, SolidWorks

EDUCATION

Worcester Polytechnic Institute, Bachelor of Science in Computer Science May 2024 | Worcester, MA

INTERESTS

Basketball, Golf, Pickleball, Hiking, Cooking, Game Development, Digital Music Creation